EE3-21
EE9-AO19

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected copy

BIOMEDICAL ELECTRONICS

Friday, 7 December 9:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer 4 out of 5 questions

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P. Georgiou
Second Marker(s) : T. Constandinou

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1. This question relates to the design of a multichannel instrument for *measuring bio-potentials from muscles.*

a) Explain how a neural action potential is generated, referring to the ionic flow of sodium and potassium ions. As part of your answer, sketch the waveform of an action potential, annotating and explaining the mechanism contributing to each phase. [4]

b) Table 1.1 lists five commonly observed bio-potentials and their amplitude and frequency characteristics. A doctor requires an instrumentation system to record from a set of electrodes connected to the back muscles shown in Figure 1.1.

Design a complete instrumentation system for recoding bio-potentials from the back according to the following specifications:

Instrumentation Amplifier Gain: 10
Low-Pass Filter Roll off: 40dB/decade
High-Pass Filter Roll off: 60dB/decade
Supply Range: -3V to 3V

State all assumptions and values of resistors and capacitors used. You may assume that the instrumentation system utilizes the whole dynamic range available and 10nF capacitors are available for the filters. [10]

c) The function of the system is to measure the muscle fiber conduction velocity (MFCV), determined by the propagation delay between the signals measured between electrodes 1-2 and 3-4. Electrode 5 is used to set the common mode voltage of the body. Design a system using your front end designed in 1.b which is able to measure the MFCV and output a digital number representing this value. You are only required to sketch the top-level architecture. State all assumptions in your design. [6]

Table 1.1

Signal	Frequency range (Hz)	Amplitude range (mV)
ECG	0.1 – 300	0.05 – 3
EEG	0.5 – 40	0.001 – 1
EMG	20 – 2000	0.001 – 100
Neural Spikes	300-5000	0.001– 0.5
Local Field Potentials	10-200	0.001 – 5

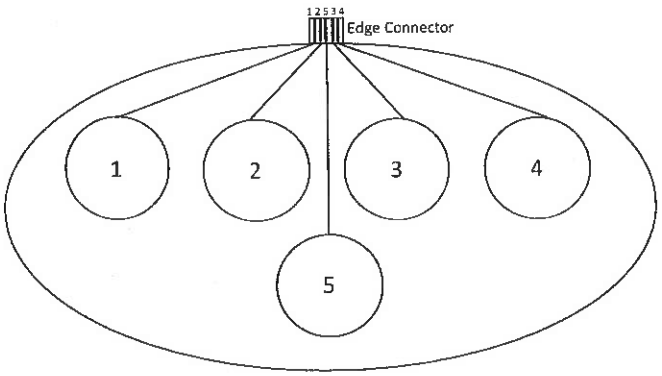


Figure 1.1

2. This question relates to *implantable neural prostheses for restoring vision*.

d) Briefly discuss the relative advantages and disadvantages of an *implantable neural prosthesis for restoring vision* that interfaces to the: (i) retina; (ii) optic nerve; (iii) visual cortex. [6]

e) Given a desired resolution of 16384 pixels (i.e. 128 x 128) is required for a retinal prosthesis, determine:

i. The area per electrode in contact with tissue (hint: you will need to estimate the total area of retina to be targeted); [2]

ii. Given the following material properties, select an electrode coating, stating your rationale; [2]

Material	Charge Density
Stainless steel	$50\mu C/cm^2$
Platinum Iridium	$150\mu C/cm^2$
Iridium Oxide	$2.5mC/cm^2$

iii. For a *current-mode, anodic-first biphasic stimulation profile*, estimate the maximum allowable stimulation current, and sketch the waveform profile; [4]

iv. Estimate the total power consumption stating any assumptions (e.g. frame rate, resolution, stimulation protocol, etc). [2]

f) Suppose that after the implantation it was established that the charge stimulus required to achieve neural activation is 10 times higher than what is achievable for the implanted electrode area and selected material, how can this be addressed (without changing the electrode material or size). [4]

3. Figure 3.1 shows an artificial pancreas system for control of blood glucose in diabetes. The system uses a continuous glucose sensor to measure a physiological glucose range of 0-20mM from interstitial fluid. The Glucose sensor comprises a working electrode (WE) with an area of $1mm^2$, a reference electrode (RE) (area:$0.5mm^2$) and a counter electrode (CE) (area:$2mm^2$). The instrumentation is implemented in CMOS and operates on a single supply with a VDD of 1.8V.

a) Sketch a top-level diagram of the closed-loop system for this artificial pancreas and describe its operation, identifying the devices in Figure 3.1 and also highlighting three main disturbances to the system.

[5]

The working electrode of the glucose sensor is coated with glucose oxidase to give a peak current to glucose at V_{Cell}=0.7V according to the following reaction:

$$\text{Glucose}+O_2 \xrightarrow{Glucose\ Oxidase} \text{gluconic acid}+ H_2O_2$$

$$H_2O_2+O_2 \xrightarrow{V=0.7V} 2H^++2e^-$$

b) Calculate the peak redox current for the reaction above using the Randels Equation below. You may assume the number of electrons n is solely given by the oxidation of H_2O_2, the scan rate is 1V/sec and the diffusion coefficient $D=4.38X10^{-8}\ m^2/s$:

$$i_p = 2.69x10^5 \cdot n^{3/2} \cdot A \cdot D^{1/2} \cdot C \cdot v^{1/2}$$

[2]

c) Design a potentiostat and trans-impedance amplifier (TIA) to instrument to the sensor making full use of the dynamic range available. Annotate the schematic and specify all component values. You may assume a 32KHz ring oscillator exists on the device to provide a digital clock.

[8]

d) Sketch a suitable circuit to provide a stable voltage to your potentiostat given that you are provided with a stable 1.2V Bandgap reference circuit on chip. You are required to calculate suitable resistor values for your voltage reference.

[5]



Figure 3.1

4. This question relates to cochlear implants for restoring hearing.

a) Discuss the key limitations to state-of-the-art implantable cochlear prostheses [5]

b) Compressed analogue (CA), Spectral Peak (SPeak) and Continuous Interleave Sampling (CIS) are three stimulation strategies that have been developed for cochlear implants. Describe in general what is the purpose of a stimulation strategy, and compare those stated above. [5]

c) Sketch a functional block diagram of a typical cochlear implant system, briefly annotating what is the function of the key blocks. [5]

d) Given a cochlear implant uses two Zinc Air batteries (each with 600mAh capacity @ 1.4V) and that the stimulator requires an average stimulation current of 200μA @ 15V, estimate the battery lifetime of the entire system stating any assumptions made. [5]

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5. This question relates to the design of an instrument for conducting DNA sequencing.

a) Explain the process of DNA sequencing describing the end-to-end procedure from an unknown strand of human DNA and the sensing mechanism used.

[10]

b) DNA sequencing has been revolutionised with the introduction of semiconductor sequencing. This uses a CMOS chip with millions of chemical sensors to conduct an in-parallel investigation of the DNA. You are required to design this CMOS chip describing the following:

- The choice of your sensor to detect DNA and the mechanism of sensing.
- The sensor operating characteristics, showing equations describing its function.
- The choice of instrumentation for this sensor.
- The circuit diagram implementation for an array to scale the design to millions of sensors.

[10]

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